

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0053

Yorkana 345 kV
August 21, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0053. The Interconnection Customer proposes a 383.2 MW solar photovoltaic facility interconnecting at the Yorkana 345 kV substation in the Pennsylvania area. The request was received on June 30, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 48 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,546,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CURECANTI-MOSER GENERATING STA 345.0 kV	TOMBSTONE-BIXBY HILL 500.0 kV	120.9	1,016	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CURECANTI-MOSER GENERATING STA 345.0 kV	Rebuild line with higher-capacity conductor	\$32,691,000	8% (\$2,546,000)
	Total allocated to project		\$2,546,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.